GRSG-DAF: A Global Robust Structured Graph Approach With Difference-Aware Filtering for SAR Image Change Detection

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***Abstract*— Synthetic aperture radar (SAR) image change detection plays a crucial role in monitoring environmental changes and landform evolution. However, existing methods often struggle to retain details while reducing computational cost, or lack the ability to effectively incorporate global information, especially under complex and noisy conditions. To address these challenges, we propose a novel global robust structured graph approach with difference-aware filtering (GRSG-DAF), which effectively enhances the detection of changed areas by combining pixel-level information with the advantages of the superpixel-based structured graph. First, we construct a structured graph by integrating difference information, utilizing superpixel co-segmentation and adaptive affinity weighting, thus significantly reducing computational complexity and preserving critical structural patterns. Second, an image reconstruction process is implemented, utilizing a feature propagation mechanism to improve the contextual representation of the image. Finally, a difference-aware guided filtering (DAGF) process is developed by integrating the inherent guidance from the original pixel-level image, preserving boundary structures and spatial details. The experimental results demonstrate that our proposed method outperforms state-of-the-art methods on five benchmark datasets, achieving higher accuracy while balancing effectiveness and efficiency in change detection.**

***Index Terms*— Change detection, difference aware, feature propagation, guided filtering, synthetic aperture radar (SAR).**

I. Introduction

CHANGE detection plays a significant role in the field of remote sensing image processing, which aims to analyze surface changes by comparing images of the same scene acquired at different times [1]. Change detection is mainly applied in fields such as agricultural and forestry monitoring [2], [3], urban expansion analysis [4], and disaster assessment [5]. Common data sources include sensors such as multispectral cameras [6], hyperspectral cameras [7], and synthetic aperture radar (SAR) [8]. In real-world scenarios, SAR is widely recognized for its significance in change detection, primarily due to its ability to capture images regardless of weather conditions. Commonly utilized SAR sensors include China's Gaofen-3 satellite, Canada's Radarsat-2 satellite, and Germany's TerraSAR satellite.

Over the past decades, mainstream research on SAR image change detection has been divided into two main types: traditional image processing methods and learning-based methods [9], [10]. Moreover, both approaches involve three sequential steps: image preprocessing, difference image (DI) generation, and DI analysis [11]. As part of SAR change detection, the image preprocessing step mainly includes operations such as radiometric correction [12], geometric correction [13], and noise suppression [14] on the original SAR images. These processes help improve image quality and reduce false changes caused by noise. In addition, image registration is critical for ensuring spatial consistency in bitemporal SAR images, as this consistency directly affects the reliability of change detection results [15]. After image preprocessing, a DI is generated by performing a differencing operation on the bitemporal SAR images [16], [17], [18]. This process aims to identify and emphasize regions where significant changes have occurred between the two acquisition times. Once the DI is generated, further analysis is conducted to extract the areas of change. Due to the potential presence of noise and false changes within the DI, methods such as cluster analysis [19] and threshold selection [20] are commonly used to classify and detect changes, ensuring the reliability of the final results. Furthermore, learning-based methods also leverage the DI to generate pseudo-labels for model training, which subsequently enables the prediction of changed pixels [21], [22]. Given their powerful feature extraction capabilities, machine learning and deep learning techniques have become central to SAR change detection research. These techniques are typically categorized into supervised [23], semi-supervised [24], and unsupervised [25] approaches. Since supervised and semi-supervised methods rely on annotated sample data and lack generalization across scenes, unsupervised approaches are the most widely applied.

Although DI generation-based methods have achieved notable success in many application scenarios, they still face limitations when dealing with highly complex and dynamically changing environments. Specifically, when the noise levels in the pre- and postimages are inconsistent, these methods often struggle to accurately distinguish changed areas, leading to increased false positive (FP) and false negative (FN) rates. In addition, although unsupervised methods avoid manual

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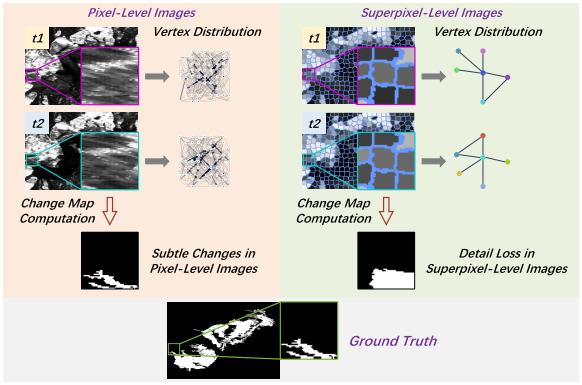


Fig. 1. Application of graph theory in SAR image change detection. For pixel-level graphs, the detection results are more detailed, but the large number of vertices results in high computational complexity. For superpixel-level graphs, the number of constructed vertices is relatively small, but this results in a loss of image details.

labels, they rely on DI clustering for pseudo-label generation, followed by pixelwise training, which results in high computational complexity. Furthermore, the interpretability of learning-based models remains limited, making it difficult to systematically improve the effectiveness of model training.

In recent years, graph theory has been increasingly applied to SAR image change detection [26], [27], [28], offering two major advantages. First, graph-based representations can effectively capture the inherent structural information of images, thus suppressing the impact of noise. Second, when constructing the graph, vertices can be defined not only as pixels but also as superpixels or patches, which helps reduce computational complexity. The application of SAR change detection based on graph theory is shown in Fig. 1. Pixel-based structured graph construction methods suffer from high computational complexity, due to the large number of vertices that need to be processed. In contrast, superpixel- or patch-based methods exhibit an insufficient correlation between vertex spatial structural features and spatial details, as well as boundary-blurring issues. These challenges make it difficult to efficiently capture pixel-level details and global semantic information. Therefore, we focus on two key questions: how to effectively extract vertex information in the structured graph and how to optimize the utilization of structured graph information while controlling computational complexity. The discussion will be conducted from the following two perspectives.

A. Key Challenges in Vertex Information Extraction

Vertex information is critical in structured graph construction, as it ensures the graph accurately captures the spatial and semantic relationships, which is essential for detecting changes and reducing noise. Existing methods for generating graph vertices either use only a single-temporal image to extract vertices and overlay them onto another temporal image [27], or rely solely on DIs to generate vertices [29]. In the field of heterogeneous change detection, it is common to co-segment bitemporal images to obtain shared vertices [30]. Nevertheless, these methods fail to establish an intrinsic connection between the spatial structural features of bitemporal images and the significant change information in the DI. As a result, vertex information becomes less sensitive to actual change regions, potentially introducing noise or overlooking important details. This undermines the accuracy of structured graph construction and distorts the representation of change information.

B. Structured Graph Construction and Global Information Integration

When constructing structured graphs, pixel-based construction methods [27], [28] preserve complete spatial details but suffer from increased computational complexity due to the large number of vertices. Patch-based methods [26] construct adjacency relationships through nonlocal similarity, reducing computational complexity. However, using fixed-size regular patches as basic units disrupts the topological structure of natural objects, leading to blurred change boundaries. In contrast, superpixel-based methods [30] can preserve object boundary structures but struggle to capture pixel-level details when solely relying on superpixel regions for graph reconstruction. The recently proposed CausalCD method [31] attempts to incorporate pixel-level information, but its graph construction is limited within individual superpixels, failing to establish global semantic associations across superpixels. Given that the boundaries of ground objects and subtle changes in SAR images are often weak, relying only on local information makes it difficult to capture critical changes accurately. Therefore, it is essential to fully utilize both pixel and superpixel information while integrating multiscale global contextual features, which helps to achieve precise extraction of change boundaries and fine details.

Given the aforementioned analysis, we propose a novel global robust structured graph approach with difference-aware filtering (GRSG-DAF), which combines the advantages of pixel-level features and superpixel-level structured graphs. By incorporating difference-aware superpixel co-segmentation, GRSG-DAF achieves more expressive vertex representations while reducing their number. To fully exploit contextual information and pixel-level details, we introduce a global feature propagation mechanism and a pixel-level guided filtering strategy. Through this framework, we aim to improve both the accuracy and computational efficiency of change detection, particularly in scenarios with high noise or low change contrast.

The contributions of this article can be summarized as follows.

1) We propose a GRSG-DAF framework, which addresses the dual objectives of preserving detailed sensitivity and reducing computational complexity in the graph-based SAR change detection process.
2) We introduce a unified structured graph that combines pixel-level guidance with superpixel-based graph modeling, followed by a two-stage refinement pipeline of global feature propagation and difference-aware guided filtering (DAGF). This cascaded design improves the discriminability of changes in the generated DI.
3) Extensive experiments on five benchmark SAR change detection datasets demonstrate that our framework achieves state-of-the-art performance in change

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detection and strikes a balance between efficiency and effectiveness.

The rest of this article is organized as follows. Section II introduces the related work. Section III provides a detailed description of the proposed GRSG-DAF. Section IV presents the experimental results along with the analysis. Section V covers the ablation studies and provides a discussion. Finally, Section VI concludes this article.

II. Related Work

In this section, we review the major research achievements of DI generation-based methods and learning-based methods in SAR change detection. Furthermore, we explore graph theory and its applications in SAR change detection.

A. DI Generation-Based Methods for SAR Image Change Detection

Early research in DI generation for SAR image change detection primarily focused on two methods: the subtraction operator [32] and the ratio operator [33]. The subtraction operator generates the DI by directly subtracting the pixels at corresponding spatial locations between two SAR images, which is simple and straightforward. In contrast, the ratio operator computes the ratio of corresponding pixels between two images, thereby partially reducing the impact of speckle noise.

To further improve change detection performance, researchers proposed the log-ratio (LR) operator [16] and the mean-ratio (MR) operator [17]. The LR operator introduces a logarithmic transformation based on the ratio operator, converting multiplicative noise into additive noise and improving the distinction between changed and unchanged areas. The MR operator computes the ratio based on the neighborhood mean of pixels, suppressing isolated noise points and yielding a denoising effect. Building on this, researchers developed the more advanced neighborhood-based ratio (NR) operator [18]. The NR operator combines pixel intensity information with spatial context from neighboring pixels, effectively suppressing speckle noise while retaining more details within the DI, thereby enhancing the reliability of change detection.

In recent years, researchers have further proposed the smoothed LR DI (SLRDI) operator [34] and the multiscale superpixel reconstruction DI (MSRDI) operator [34], aiming to address the limitations of previous operators. The SLRDI operator applies smoothing to the LR operator, effectively reducing noise in SAR images while preserving finer details. The MSRDI operator employs a multiscale superpixel reconstruction method [35] to extract image features at different scales, resulting in significant noise reduction. By integrating the strengths of various methods, these advanced techniques significantly improve the effectiveness of SAR change detection. Once the DI is generated, postprocessing methods such as Otsu thresholding [36], fuzzy c-means (FCM) [37], FLICM [38], PCAK [39], and conditional random field (CRF) [40], [41] are applied to obtain the final change map.

B. Learning-Based Methods for SAR Image Change Detection

Thanks to their robust feature representation capabilities, learning-based techniques have been widely adopted in remote sensing, achieving remarkable performance across various domains, including object detection [42], moving object detection [43], and object classification [44]. In the field of SAR change detection, learning-based approaches have gradually become a key research focus and can be categorized into supervised, semi-supervised, and unsupervised methods [22]. Specifically, supervised and semi-supervised methods utilize annotated data to provide prior information, leading to superior detection performance. Wang et al. [23] proposed a supervised lightweight network for bitemporal SAR change detection, employing bottleneck layers and dilated convolutional kernels to reduce computational complexity while improving generalization performance in complex scenes. Hou et al. [24] introduced a deep collaborative semi-supervised learning framework with class-rebalancing for SAR change detection, which dynamically optimizes deep features and clustering-based pseudo-labels, achieving high detection accuracy with a small amount of annotated data. However, in practical applications, the availability of annotated samples is extremely limited, restricting the development of supervised methods. Additionally, semi-supervised methods require task-specific model training, limiting their generalizability to novel data distributions and application scenarios.

Unsupervised learning-based methods, which obviate the need for annotated data, have demonstrated superior applicability in SAR image change detection. Qu et al. [11] proposed a dual-domain network (DDNet) that, for the first time, integrates discrete cosine transform-domain features with spatial-domain features. They designed a multiregion convolutional module to enhance the central region features in SAR image patches, thereby effectively suppressing speckle noise. Meng et al. [25] proposed a layer attention-based noise-tolerant network (LANTNet), which dynamically fuses multilayer convolutional features through a layer attention module. They developed a noise-resistant loss function to mitigate the impact of pseudo-label noise during model training, improving feature discriminability. Zhang et al. [45] introduced a convolution and attention mixer (CAMixer), which fuses shifted convolution and self-attention mechanisms in a parallel structure. They also employed a gated feedforward network to enhance nonlinear feature representations, balancing both local details and global dependencies in SAR images. Xie et al. [46] proposed a wavelet-based bidimensional aggregation network (WBANet), employing discrete wavelet transform (DWT) and inverse DWT (IDWT) for downsampling while preserving the high-frequency information of SAR images. Dong et al. [47] proposed a weighted contrastive learning framework that captures complex contextual information through global channel aggregation and multiscale fusion, enabling SAR change detection across multiple resolutions. Duan et al. [48] utilized DWT to separate high-frequency noise and employed a focal loss function to enhance the change detection performance in SAR images under varying noise

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conditions. Chen et al. [49] introduced a shallow broad wavelet scattering network (SWSN), which extracts frequency-domain features. They developed a dual-domain framework that integrates neighborhood ratio features from bitemporal images, which enhances noise suppression and reduces computational complexity.

C. Graph-Based Perspective on SAR Image Change Detection

Graph theory [50], [51], which describes the spatial relationships between pixels in images, has been widely applied in SAR image processing tasks such as segmentation [52], denoising [53], recognition [54], and classification [55]. These graph-based approaches leverage structural dependencies, nonlocal self-similarity, and adaptive attention mechanisms to improve robustness against speckle noise, limited samples, and complex scene variations in SAR images.

Beyond applications in other fields, graph theory has also found increasing use in change detection, particularly for SAR data [28], [56], [57], optical data [58], [59], [60], and heterogeneous data [30], [61], [62]. Pham et al. [56] proposed a change detection method based on keypoints and graph theory. Their approach constructs a structured graph to compare the local geometric consistency of bitemporal SAR images, thereby improving detection accuracy and reducing computational complexity. Sun et al. [57] introduced an improved nonlocal patch-based graph (INLPG) method, which detects changes by comparing structural consistency in bitemporal SAR images rather than directly comparing pixel values. This approach is robust to noise, lighting, and other disturbances. Wang et al. [27] proposed an SAR change detection method based on a hierarchical heterogeneous graph, which leverages multiscale features and heterogeneous affinity to improve change detection accuracy. Li et al. [31] proposed a graph contrastive learning framework based on causal inference. Their framework reduces the interference of pseudo-label noise by incorporating a structural causal model and causal intervention strategy. Zhu et al. [63] proposed an unsupervised spatial self-similarity difference-based change detection (USSD) method, which constructs a structure similar to a patch-similarity graph to fuse amplitude and similarity features, effectively suppressing SAR image noise and enhancing the distinction of changed areas. Li et al. [29] developed a structure-optimized complex-valued graph contrastive learning model, which integrates a graph reconstruction strategy optimized by spectral clustering. This strategy achieves a significant improvement in change detection performance under limited annotated data conditions. Wang et al. [28] proposed a heterogeneous graph with multilevel and multiorder connections (M2HG) that captures local, nonlocal, and global information in SAR images. They subsequently used a composite random walk matrix to quantify these connections, yielding an enhanced ability to distinguish changed areas. In this study, we develop a graph-based method for change detection in SAR images, which is designed to robustly capture and propagate both local and global contextual features under noisy and complex conditions.

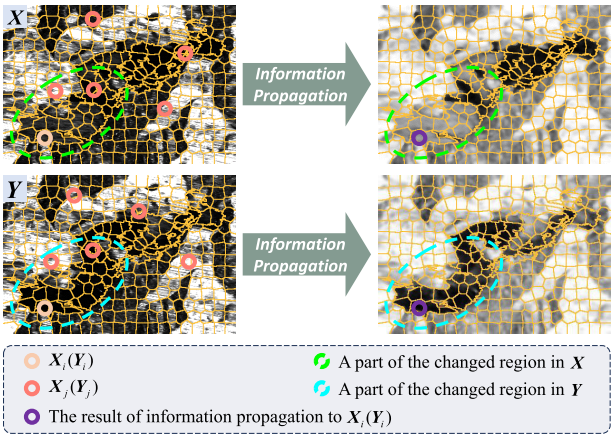


Fig. 2. Neighborhood information propagation on the GRSG. For $\boldsymbol{X}_i$ and $\boldsymbol{Y}_i$ in the same region, their global neighbors $\boldsymbol{X}_j$ and $\boldsymbol{Y}_j$ are located across the entire image, and the neighbor information is propagated to $\boldsymbol{X}_i$ and $\boldsymbol{Y}_i$.

III. Methodology

The SAR images acquired at two different times (t_1 and t_2) are defined as $\boldsymbol{X} \in \mathbb{R}^{M\times N}$ and $\boldsymbol{Y} \in \mathbb{R}^{M\times N}$, with the corresponding pixels represented as $x(m,n)$ and $y(m,n)$, respectively. According to image self-similarity [26], each small region $\boldsymbol{X}_i$ in the image can always find a similar region $\boldsymbol{X}_j$. As shown in Fig. 2, the GRSG is defined as follows: for each $\boldsymbol{X}_i$ ($\boldsymbol{Y}_i$), its global neighbors $\boldsymbol{X}_j$ ($\boldsymbol{Y}_j$) are located so that the information from $\boldsymbol{X}_j$ ($\boldsymbol{Y}_j$) can be propagated to $\boldsymbol{X}_i$ ($\boldsymbol{Y}_i$), thereby enhancing regions with subtle changes. As illustrated in Fig. 3, this study employs a GRSG with global neighborhood information propagation and DAGF. The method involves three main steps: 1) extracting a difference-guided superpixel co-segmentation map and constructing a GRSG using superpixel-based vertices; 2) propagating neighborhood information based on the structured graph to reconstruct images from two different times; and 3) applying DAGF to optimize the pixel-level details of the reconstructed images for obtaining the final change map.

A. Difference-Guided GRSG

Leveraging various methods for calculating DIs in SAR change detection, such as the LR, MR, and NR operators, we extract the initial DI (IDI) using the LR operator. Based on the difference information contained in the IDI, a pseudo-color image $\mathcal{P} = [\boldsymbol{X}, \boldsymbol{Y}] \oplus [\text{IDI}]$ is constructed, where $\oplus$ represents the synthesis operation. In this context, the SAR images at times t_1 and t_2 (denoted as $\boldsymbol{X}$ and $\boldsymbol{Y}$) are used as the first two channels of $\mathcal{P}$, while the third channel can be IDI or a zero channel (where IDI is separately fused with the first two channels). To facilitate the utilization of difference information in conjunction with $\boldsymbol{X}$ and $\boldsymbol{Y}$ during the processing of $\mathcal{P}$, we choose IDI as the third channel. Then, the simple linear iterative clustering (SLIC) [35] method is applied to generate superpixels from $\mathcal{P}$.

Once $\mathcal{P}$ is co-segmented into N_S superpixels, we obtain the segmentation map Λ, which contains difference information. Subsequently, Λ can be used to segment $\boldsymbol{X}$ and $\boldsymbol{Y}$ into superpixels $\boldsymbol{X}_i$ and $\boldsymbol{Y}_i$, $i = 1, 2, \ldots, N_S$, with identical

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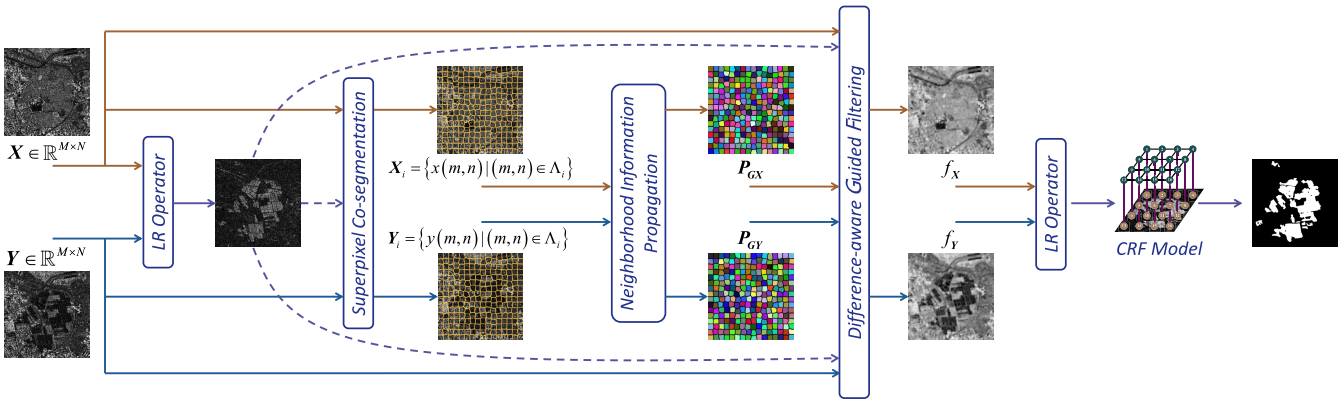


Fig. 3. Framework of the proposed GRSG-DAF method. The core method consists of three parts: superpixel co-segmentation, neighborhood information propagation, and DAGF. The LR operator, which stands for the LR operator, is first used to compute the difference information between the bitemporal inputs. It is, then, applied to the outputs of DAGF, f_X and f_Y, with the obtained difference results subsequently segmented using a CRF to generate the final change map.

contours, defined as

$$\boldsymbol{X}_i = \{x(m,n) \mid (m,n) \in \Lambda_i\} \tag{1}$$

$$\boldsymbol{Y}_i = \{y(m,n) \mid (m,n) \in \Lambda_i\} \tag{2}$$

where $\boldsymbol{X}_i$ and $\boldsymbol{Y}_i$ represent the same geographical region, with each containing similar features. Notably, by incorporating difference information from different times, the co-segmentation process makes the generated superpixels more sensitive to changes. Since the number of pixels within each superpixel of $\boldsymbol{X}_i$ and $\boldsymbol{Y}_i$ varies, it is necessary to extract features of the same dimensionality when constructing the structured graph, yielding feature matrices $\boldsymbol{X}'$ and $\boldsymbol{Y}'$. To facilitate subsequent information propagation, 1-D mean features are used in this study.

Once superpixel-level features are extracted, each superpixel is treated as a vertex, and a structured graph is constructed to represent the global neighborhood information of each vertex. For the image $\boldsymbol{X}$ at time t_1, the graph $G^{t1} = \{\mathcal{V}^{t1}, \mathcal{E}^{t1}, \boldsymbol{A}^{t1}\}$ is constructed as

$$\mathcal{V}^{t1} = \{\boldsymbol{X}_i \mid i = 1, 2, \ldots, N_S\} \tag{3}$$

$$\mathcal{E}^{t1} = \left\{\left(\boldsymbol{X}_i, \boldsymbol{X}_j\right) \mid i = 1, 2, \ldots, N_S, j \in \mathcal{N}_i^x\right\} \tag{4}$$

$$A_{i,j}^{t1} = \omega_{\text{sp}}\left(\boldsymbol{X}_i, \boldsymbol{X}_j\right) \ \forall\left(\boldsymbol{X}_i, \boldsymbol{X}_j\right) \in \mathcal{E}^{t1} \tag{5}$$

where the global neighbor set $\mathcal{N}_i^x$ for superpixel $\boldsymbol{X}_i$ is constructed by selecting the top k minimum values (excluding i) from the feature difference vector $\boldsymbol{D}_i^x$. Specifically, $\boldsymbol{D}^x$ is the global feature difference matrix of image $\boldsymbol{X}$, with each element defined as $d_{i,j}^x = \|\boldsymbol{X}_i' - \boldsymbol{X}_j'\|_2^2$, where $\boldsymbol{X}_i'$ and $\boldsymbol{X}_j'$ are the feature representations of superpixels $\boldsymbol{X}_i$ and $\boldsymbol{X}_j$, respectively. The adaptive affinity-weighted adjacency matrix $\boldsymbol{A}^{t1}$ is defined as $A_{i,j}^{t1} = 0$ when $(\boldsymbol{X}_i, \boldsymbol{X}_j) \notin \mathcal{E}^{t1}$, and when $(\boldsymbol{X}_i, \boldsymbol{X}_j) \in \mathcal{E}^{t1}$, the weight ω_{sp} between $\boldsymbol{X}_i$ and $\boldsymbol{X}_j$ is calculated based on feature affinity

$$\omega_{\text{sp}}\left(\boldsymbol{X}_i, \boldsymbol{X}_j\right) = \exp\left\{-\sum \log\left(\frac{\boldsymbol{X}_i' + \varepsilon}{\boldsymbol{X}_j' + \varepsilon} + \frac{\boldsymbol{X}_j' + \varepsilon}{\boldsymbol{X}_i' + \varepsilon}\right)\right\}. \tag{6}$$

The weight ω_{sp} is adaptively adjusted based on the feature affinity between $\boldsymbol{X}_i$ and its global neighbors in $\mathcal{N}_i^x$, ensuring that the affinity measure between superpixels is smooth and stable, which is essential for handling noisy SAR images.

Similarly, for the image $\boldsymbol{Y}$ at time t_2, its feature differences $d_{i,j}^y = \|\boldsymbol{Y}_i' - \boldsymbol{Y}_j'\|_2^2$ and difference matrix $\boldsymbol{D}^y$ are defined, and the structured graph $G^{t2} = \{\mathcal{V}^{t2}, \mathcal{E}^{t2}, \boldsymbol{A}^{t2}\}$ is constructed following the same steps:

$$\mathcal{V}^{t2} = \{\boldsymbol{Y}_i \mid i = 1, 2, \ldots, N_S\} \tag{7}$$

$$\mathcal{E}^{t2} = \left\{\left(\boldsymbol{Y}_i, \boldsymbol{Y}_j\right) \mid i = 1, 2, \ldots, N_S, j \in \mathcal{N}_i^y\right\} \tag{8}$$

$$A_{i,j}^{t2} = \omega_{\text{sp}}\left(\boldsymbol{Y}_i, \boldsymbol{Y}_j\right) \ \forall\left(\boldsymbol{Y}_i, \boldsymbol{Y}_j\right) \in \mathcal{E}^{t2} \tag{9}$$

where $\mathcal{N}_i^y$ represents the global neighbor set for superpixel $\boldsymbol{Y}_i$, and $\boldsymbol{A}^{t2}$ represents the adaptive adjacency matrix for $\boldsymbol{Y}$.

B. Image Reconstruction via Structured Graph Global Neighborhood Information Propagation

In the constructed structured graph, low-weight connections within ω_{sp} are removed. Then, each superpixel is used as an ego-vertex to create an ego graph, resulting in N_S noise-robust ego graphs. These ego graphs connect superpixels with significant affinity, ensuring the robustness of the graph structure. Among the collection of all N_S graph representations, a message-passing mechanism is applied to each ego graph, propagating information from global neighbors to the current superpixel vertex. As shown in Fig. 4, the principle of global information propagation is as follows: the feature means of each superpixel ($\overline{\boldsymbol{X}_i}$ and $\overline{\boldsymbol{Y}_i}$) is aggregated with the feature means of their global neighbors ($\overline{\boldsymbol{X}_j}$ and $\overline{\boldsymbol{Y}_j}$) through an energy minimization framework, thereby amplifying features among strongly connected regions. Specifically, the new feature representations $\boldsymbol{P}_{\boldsymbol{GX}_i}$ and $\boldsymbol{P}_{\boldsymbol{GY}_i}$ ($i = 1, 2, \ldots, N_S$) are obtained by minimizing the weighted discrepancy between local (self) features and global neighbor features, formulated as

$$\boldsymbol{P}_{\boldsymbol{GX}_i} = \arg\min_{\boldsymbol{P}} \left\|\boldsymbol{P} - \overline{\boldsymbol{X}_i}\right\|_2^2 + \sum_{j \in \mathcal{N}_i^x} A_{i,j}^{t1} \left\|\boldsymbol{P} - \overline{\boldsymbol{X}_j}\right\|_2^2 \tag{10}$$

$$\boldsymbol{P}_{\boldsymbol{GY}_i} = \arg\min_{\boldsymbol{P}} \left\|\boldsymbol{P} - \overline{\boldsymbol{Y}_i}\right\|_2^2 + \sum_{j \in \mathcal{N}_i^y} A_{i,j}^{t2} \left\|\boldsymbol{P} - \overline{\boldsymbol{Y}_j}\right\|_2^2 \tag{11}$$

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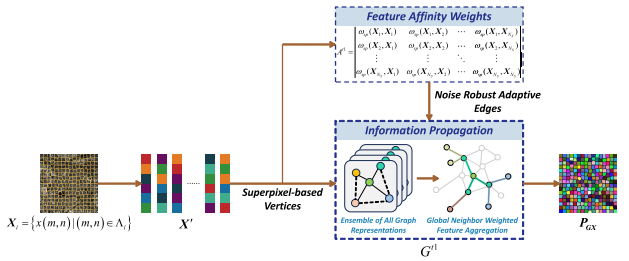


Fig. 4. Image reconstruction of $\boldsymbol{X}$ based on global neighborhood information propagation. The dashed lines in an ego graph represent edges that belong to other ego graphs. For each vertex $\boldsymbol{X}_i$, information is propagated and updated based on the feature affinity weights in the ego graph.

where the adaptive edge weights $A_{i,j}^{t1}$ and $A_{i,j}^{t2}$ represent the affinity between vertex pairs, with higher weights indicating stronger affinity. The closed-form solutions to (10) and (11) are equivalent to weighted averaging of local and global features, indicating that global neighbors with higher weights will have a more significant influence on the current vertex during the propagation process. The unified formulation is expressed as

$$\boldsymbol{P}_{\boldsymbol{G}\boldsymbol{\phi}_i} = \frac{\overline{\boldsymbol{\phi}_i} + \sum_{j\in\mathcal{N}_i^{\phi}} \overline{\boldsymbol{\phi}_j} A_{i,j}^{\tau(\phi)}}{1 + \left\| \boldsymbol{A}^{\tau(\phi)}\left[i, \mathcal{N}_i^{\phi}\right]\right\|_1} \quad (\phi \in \{\boldsymbol{X}, \boldsymbol{Y}\}) \tag{12}$$

where $\|\cdot\|_1$ restricted to $\mathcal{N}_i^{\phi}$ is defined as $\|\boldsymbol{A}[i,S]\|_1 := \sum_{j\in S}|A_{i,j}|$, with $\tau(\boldsymbol{X}) = t1$ and $\tau(\boldsymbol{Y}) = t2$ determining the exponents for the edge weights. Here, $\mathcal{N}_i^{\phi}$ equals $\mathcal{N}_i^{x}$ when $\phi = \boldsymbol{X}$ and $\mathcal{N}_i^{y}$ when $\phi = \boldsymbol{Y}$.

The calculated $\boldsymbol{P}_{\boldsymbol{G}\boldsymbol{X}_i}$ and $\boldsymbol{P}_{\boldsymbol{G}\boldsymbol{Y}_i}$ values are then assigned to all pixels within their corresponding superpixel regions, yielding the reconstructed results $\boldsymbol{P}_{\boldsymbol{G}\boldsymbol{X}}$ and $\boldsymbol{P}_{\boldsymbol{G}\boldsymbol{Y}}$

$$\boldsymbol{P}_{\boldsymbol{G}\boldsymbol{X}}(m,n) = \sum_{i=1}^{N_S} \boldsymbol{P}_{\boldsymbol{G}\boldsymbol{X}_i} \cdot \mathbf{1}_{\Lambda_i}(m,n) \tag{13}$$

$$\boldsymbol{P}_{\boldsymbol{G}\boldsymbol{Y}}(m,n) = \sum_{i=1}^{N_S} \boldsymbol{P}_{\boldsymbol{G}\boldsymbol{Y}_i} \cdot \mathbf{1}_{\Lambda_i}(m,n) \tag{14}$$

where $\mathbf{1}_{\Lambda_i}(m,n)$ is an indicator function defined as: $\mathbf{1}_{\Lambda_i}(m,n) = 1$ if (m,n) belongs to superpixel region Λ_i; otherwise, it equals 0. These reconstructed results, $\boldsymbol{P}_{\boldsymbol{G}\boldsymbol{X}}$ and $\boldsymbol{P}_{\boldsymbol{G}\boldsymbol{Y}}$, integrate superpixel-level features with global features through the information propagation across the global neighborhood. Consequently, when aggregating global neighborhood information, the change-related features are further enhanced, allowing for a more accurate and comprehensive representation of the actual changes in the original images.

C. Image Detail Refinement Using DAGF

When superpixels are used as vertices in the reconstruction of images $\boldsymbol{X}$ and $\boldsymbol{Y}$, spatial details within the superpixel blocks are inevitably lost. To enhance the semantic information of the reconstructed images, we employ $\boldsymbol{X}$ and $\boldsymbol{Y}$ as guide images and apply guided filtering to the input images $\boldsymbol{P}_{\boldsymbol{G}\boldsymbol{X}}$ and $\boldsymbol{P}_{\boldsymbol{G}\boldsymbol{Y}}$. As shown in Fig. 5, guided filtering leverages the structural features of the guide images, thereby ensuring that critical boundary structures and spatial details are preserved after the image reconstruction process.

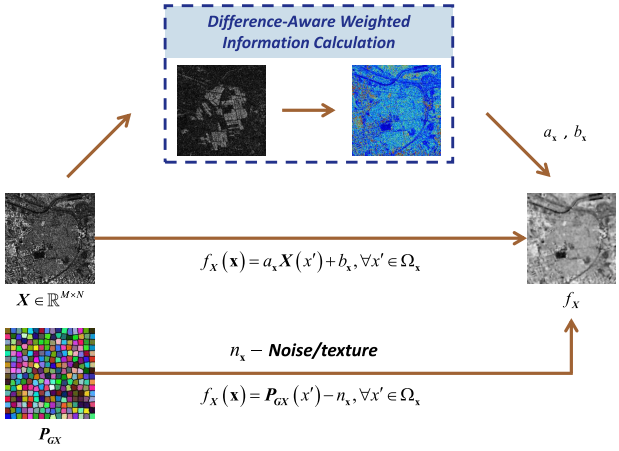


Fig. 5. Guided filtering for $\boldsymbol{P}_{\boldsymbol{G}\boldsymbol{X}}$ using information from the original image $\boldsymbol{X}$ and IDI.

In the guided filtering process, a regularization term is added to preserve the gradient of the guide image [64]. For the change detection task in this study, to focus on regions prone to changes during guiding, a difference-aware weighting operator is introduced into the regularization term

$$\Gamma_{\mathrm{IDI}}(di') = \frac{1}{MN}\sum_{dj'=1}^{MN} \frac{\sigma^2(\mathrm{IDI} \mid \Lambda_{\mathcal{M}(di')}) + \eta}{\sigma^2(\mathrm{IDI} \mid \Lambda_{\mathcal{M}(dj')}) + \eta} \tag{15}$$

where $\mathcal{M}(\cdot)$ is a mapping function such that $\mathcal{M}(di')$ denotes the position of pixel di' in the superpixel of the segmentation map Λ. IDI $\mid \Lambda_{\mathcal{M}(di')}$ represents all pixels in IDI belonging to the superpixel $\Lambda_{\mathcal{M}(di')}$.

For superpixels with significant detail changes in the IDI, the value of Γ_{IDI} tends to be larger, meaning that the details in images $\boldsymbol{X}$ and $\boldsymbol{Y}$ should be preserved. With respect to image $\boldsymbol{X}$, the loss function for guided filtering is constructed as

$$\mathcal{J}(a_{\mathbf{x}}, b_{\mathbf{x}}) = \sum_{x'\in\Omega_{\mathbf{x}}} \left(\left(a_{\mathbf{x}}\boldsymbol{X}(x') + b_{\mathbf{x}} - \boldsymbol{P}_{\boldsymbol{G}\boldsymbol{X}}(x')\right)^2 + \frac{\epsilon}{\Gamma_{\mathrm{IDI}}(x')} a_{\mathbf{x}}^2 \right) \tag{16}$$

where $\Omega_{\mathbf{x}}$ is the window surrounding pixel x'. By solving for the parameters that minimize (16), we obtain

$$a_{\mathbf{x}} = \frac{\mathrm{Cov}_{\Omega_x}[\boldsymbol{X}(\mathbf{x}), \boldsymbol{P}_{\boldsymbol{G}\boldsymbol{X}}(\mathbf{x})]}{\mathrm{Var}_{\Omega_x}[\boldsymbol{X}(\mathbf{x})] + E_{\Omega_x}\left[\epsilon / \Gamma_{\mathrm{IDI}}(\mathbf{x})\right]} \tag{17}$$

$$b_{\mathbf{x}} = E_{\Omega_{\mathbf{x}}}[\boldsymbol{P}_{\boldsymbol{G}\boldsymbol{X}}(\mathbf{x})] - a_{\mathbf{x}} E_{\Omega_{\mathbf{x}}}[\boldsymbol{X}(\mathbf{x})]. \tag{18}$$

The derivation process of $a_{\mathbf{x}}$ and $b_{\mathbf{x}}$ is provided in the Appendix. For regions within $\Omega_{\mathbf{x}}$ that contain more changing superpixels, $E_{\Omega_x}[\Gamma_{\mathrm{IDI}}(\mathbf{x})]$ tends to be larger, resulting in a larger $a_{\mathbf{x}}$ and improved preservation of spatial details.

Finally, the guided image f_X for image $\boldsymbol{X}$ can be obtained as

$$f_X(\mathbf{x}) = \frac{1}{|\Omega_{\mathbf{x}'}|} \sum_{\mathbf{x}':\mathbf{x}\in\Omega_{\mathbf{x}'}} (a_{\mathbf{x}'}\boldsymbol{X}(\mathbf{x}) + b_{\mathbf{x}'}). \tag{19}$$

Similarly, the guided image f_Y for image $\boldsymbol{Y}$ is obtained as

$$f_Y(\mathbf{y}) = \frac{1}{|\Omega_{\mathbf{y}'}|} \sum_{\mathbf{y}':\mathbf{y}\in\Omega_{\mathbf{y}'}} \left(a_{\mathbf{y}'}\boldsymbol{Y}(\mathbf{y}) + b_{\mathbf{y}'}\right). \tag{20}$$

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Algorithm 1 Procedure of GRSG-DAF

Input: Bitemporal image pairs $\boldsymbol{X}$ and $\boldsymbol{Y}$
Output: Final change map
Structured graph construction:
  Generate superpixels from $\boldsymbol{X}$ and $\boldsymbol{Y}$.
  Construct the segmentation map Λ from superpixels.
  Construct the graphs $G^{t1} = \{\mathcal{V}^{t1}, \mathcal{E}^{t1}, \boldsymbol{A}^{t1}\}$ with Eqn (3), (4), (5), and $G^{t2} = \{\mathcal{V}^{t2}, \mathcal{E}^{t2}, \boldsymbol{A}^{t2}\}$ with Eqn (7), (8), (9).
Global information propagation:
  for each $\boldsymbol{X}_i$ and $\boldsymbol{Y}_i$:
    Propagate information to $\boldsymbol{X}_i$ through Eqn (10).
    Propagate information to $\boldsymbol{Y}_i$ through Eqn (11).
  Assign the new values to the corresponding pixels in $\boldsymbol{P}_{GX}$ and $\boldsymbol{P}_{GY}$ through Eqn (13), (14).
Guided filtering:
  Apply guided filtering using $\boldsymbol{X}, \boldsymbol{Y}$, and IDI as guides for $\boldsymbol{P}_{GX}, \boldsymbol{P}_{GY}$, obtaining f_X and f_Y.
Change extraction:
  Compute $\mathrm{DI}_{\mathrm{final}}$ with Eqn (21).
  Segment $\mathrm{DI}_{\mathrm{final}}$ to extract the binary change map.

After obtaining the guided images f_X and f_Y, the LR operator can still be used to generate the final DI

$$\mathrm{DI}_{\mathrm{final}} = |\log(f_X./f_Y)|. \tag{21}$$

Once the $\mathrm{DI}_{\mathrm{final}}$ is obtained, postprocessing can be performed using methods such as Otsu thresholding [36], FCM [37], FLICM [38], PCAK [39], and CRF [40], [41] to obtain the final change map. We utilize a CRF approach based on iterated conditional modes [40] for postprocessing, which can capture spatial relationships between pixels to achieve smoother segmentation results. Specifically, we apply k-means to initially segment the DI into changed and unchanged regions, followed by two iterations of label refinement by minimizing a total energy function. This function integrates feature field and label field energy, where the latter promotes spatial smoothness through a second-order eight-connected neighborhood system using a fixed potential parameter of 1 [40].

The overall framework of the SAR image change detection method proposed in this study is summarized in Algorithm 1. The time complexity of the algorithm is primarily determined by superpixel co-segmentation, structured graph construction, and guided filtering. For superpixel co-segmentation, the SLIC algorithm is employed with a time complexity of $\mathcal{O}(MN)$. In the structured graph construction, the time complexity of finding the neighbors for each vertex is $\mathcal{O}(N_S^2)$. For guided filtering, due to the use of the integral image [64], the time complexity is approximately linear, $\mathcal{O}(MN)$. For this reason, the time complexity of the proposed GRSG-DAF is mainly determined by the number of superpixels N_S, indicating the high efficiency of superpixel-based graph construction.

IV. Experimental Results and Analysis

A. Dataset Description

To better evaluate the effectiveness of the proposed GRSG-DAF method, experiments were conducted on five SAR change detection datasets [23], [65]. Each dataset was carefully selected to cover a variety of sensors, thereby ensuring a comprehensive evaluation of the proposed GRSG-DAF under different imaging conditions. The image sizes across different datasets range from 275 to 641 pixels in length or width. Additionally, the number of changed pixels ranges from 1348 to 59 052, reflecting significant differences in the size of changed areas. Furthermore, the datasets include various change events, such as floods and earthquakes. These events cause different degrees of impact on the backscatter intensity of SAR images, which allows for a better assessment of the generalization ability of the proposed GRSG-DAF. Table I summarizes the datasets used in this study, detailing the characteristics of each dataset. These datasets are displayed in Figs. 6–10, where the ground truth was meticulously annotated by experts with relevant knowledge.

B. Evaluation Metrics

The method proposed in this study falls under the category of binary change detection, which is essentially a binary classification task. Therefore, we evaluate the method based on five standard metrics used in binary classification tasks: FP, which refers to the number of unchanged areas that are incorrectly classified as changed areas; FN, which refers to the number of changed areas that are incorrectly classified as unchanged areas; overall error (OE), which refers to the total number of misclassified samples; percentage correct classification (PCC), which refers to the proportion of samples that are correctly classified by the model overall; and the Kappa coefficient (KC), which is used to evaluate the classification accuracy and reliability of the model. Additionally, we introduce a runtime metric to quantitatively compare the computational efficiency of the various methods. The formulas for calculating PCC and KC are as follows:

$$\mathrm{PCC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{22}$$

$$\mathrm{KC} = \frac{\mathrm{PCC} - \mathrm{PRE}}{1 - \mathrm{PRE}} \tag{23}$$

$$\mathrm{PRE} = \frac{(\mathrm{TP} + \mathrm{FN})(\mathrm{TP} + \mathrm{FP}) + (\mathrm{TN} + \mathrm{FP})(\mathrm{TN} + \mathrm{FN})}{(\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN})^2} \tag{24}$$

where TP reflects the number of correct classifications of positive samples by the model, and TN reflects the number of correct classifications of negative samples by the model. The metrics mentioned above are widely recognized in SAR change detection research.

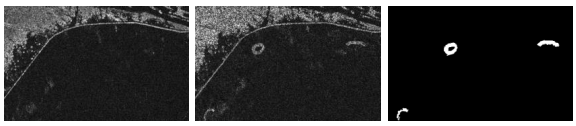
C. Experimental Setting

For each dataset, the number of neighbors used is 9 when constructing an ego graph. For smaller datasets like Datasets #1, #2, and #5, the number of superpixels is 3000 for Datasets #1 and #2, and 1000 for Dataset #5. For larger datasets, such as Datasets #3 and #4, the number of superpixels is 7000.

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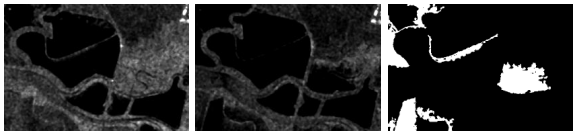
TABLE I
DESCRIPTION OF THE CHANGE DETECTION DATASETS UTILIZED

Dataset	Location	Event	Pre-Date	Post-Date	Size	Changed Pixels	Sensor
#1	Yellow River, China	Coastline Changes	18 June 2008	19 June 2009	280 × 450	1,348	Radarsat-2
#2	San Francisco, USA	Flooding	10 August 2003	16 May 2004	275 × 400	13,969	ERS-2
#3	Sendai, Japan	Earthquake	20 October 2010	6 May 2011	560 × 549	59,052	TerraSAR-X
#4	Sendai, Japan	Earthquake	20 October 2010	6 May 2011	641 × 613	38,411	TerraSAR-X
#5	Wenchuan, China	Earthquake	3 March 2008	16 June 2008	301 × 442	22,940	ESA/ASAR



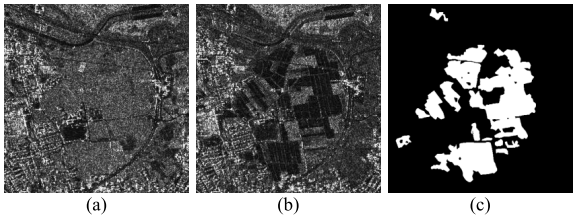
(a) (b) (c)

Fig. 6. Dataset #1. (a) Image obtained in June 2008. (b) Image obtained in June 2009. (c) Ground truth.



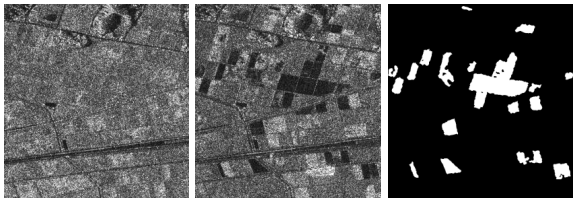
(a) (b) (c)

Fig. 7. Dataset #2. (a) Image obtained in August 2003. (b) Image obtained in May 2004. (c) Ground truth.



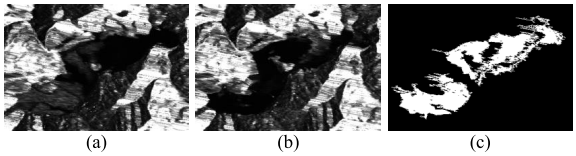
(a) (b) (c)

Fig. 8. Dataset #3. (a) Image obtained in October 2010. (b) Image obtained in May 2011. (c) Ground truth.



(a) (b) (c)

Fig. 9. Dataset #4. (a) Image obtained in October 2010. (b) Image obtained in May 2011. (c) Ground truth.



(a) (b) (c)

Fig. 10. Dataset #5. (a) Image obtained in March 2008. (b) Image obtained in June 2008. (c) Ground truth.

Section V will provide an in-depth analysis of the sensitivity of these parameters, illustrating their impact on the method's performance.

TABLE II
QUANTITATIVE EVALUATION OF DIFFERENT CHANGE DETECTION METHODS ON DATASET #1. BEST RESULTS ARE BOLDED, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Method	FP	FN	OE	PCC(%)	KC(%)	Time(s)
DDNet [11]	6,442	63	6,505	94.84	26.99	573.31
INLPG [57]	**73**	293	<u>366</u>	<u>99.71</u>	<u>85.07</u>	25.44
LANTNet [25]	6,315	47	6,362	94.95	27.71	361.74
CAMixer [45]	5,198	54	5,252	95.83	31.79	101.38
WBANet [46]	7,710	49	7,759	93.84	23.66	105.05
USSD [63]	3,296	180	3,476	97.24	39.19	<u>8.43</u>
M2HG [28]	24,127	<u>23</u>	24,150	80.83	8.02	645.61
SWSN [49]	24,932	**0**	24,932	80.21	7.88	340.89
GRSG-DAF	<u>201</u>	142	**343**	**99.73**	**87.41**	**8.42**

To reveal the superiority of the proposed GRSG-DAF, this section presents a comparison with the results of eight state-of-the-art methods, including DDNet[1] [11], INLPG[2] [57], LANTNet [25], CAMixer [45], WBANet[3] [46], USSD[4] [63], M2HG[5] [28], and SWSN[6] [49]. All experiments are performed on a standardized hardware configuration (an NVIDIA GeForce RTX 3080 GPU alongside a 12th Gen Intel Core i7-12700KF 12-Core CPU with 32-GB RAM) to ensure fairness. DDNet, LANTNet, CAMixer, and WBANet are implemented in PyTorch, whereas the remaining methods are developed using MWorks, which is consistent with the authors' open-source implementations.

D. Experimental Results and Comparisons

Tables II–VI provide quantitative comparison results, while Fig. 11 presents visual comparison results. To better highlight the locations of error regions in detection, FPs are marked in green and FNs are marked in magenta, while true positives and true negatives are depicted in white and black, respectively.

1) Results on Dataset #1: Dataset #1 primarily showcases changes along the water–land boundary, where noise within the land area affects the detection results. The visual comparison of various methods is presented in the first row of Fig. 11, and the corresponding quantitative results are illustrated in Table II. It can be observed that DDNet, LANTNet, CAMixer, WBANet, USSD, M2HG, and SWSN exhibit relatively high PCC values, all exceeding 80%. However, due

[1]https://github.com/summitgao
[2]https://github.com/yulisun/INLPG
[3]https://github.com/xuwayyy/WBANet
[4]https://github.com/zhuzhu94854693/USSD
[5]https://github.com/JunWangZJ/M2HG_CD
[6]https://github.com/Jade-ChenGong/SWSN

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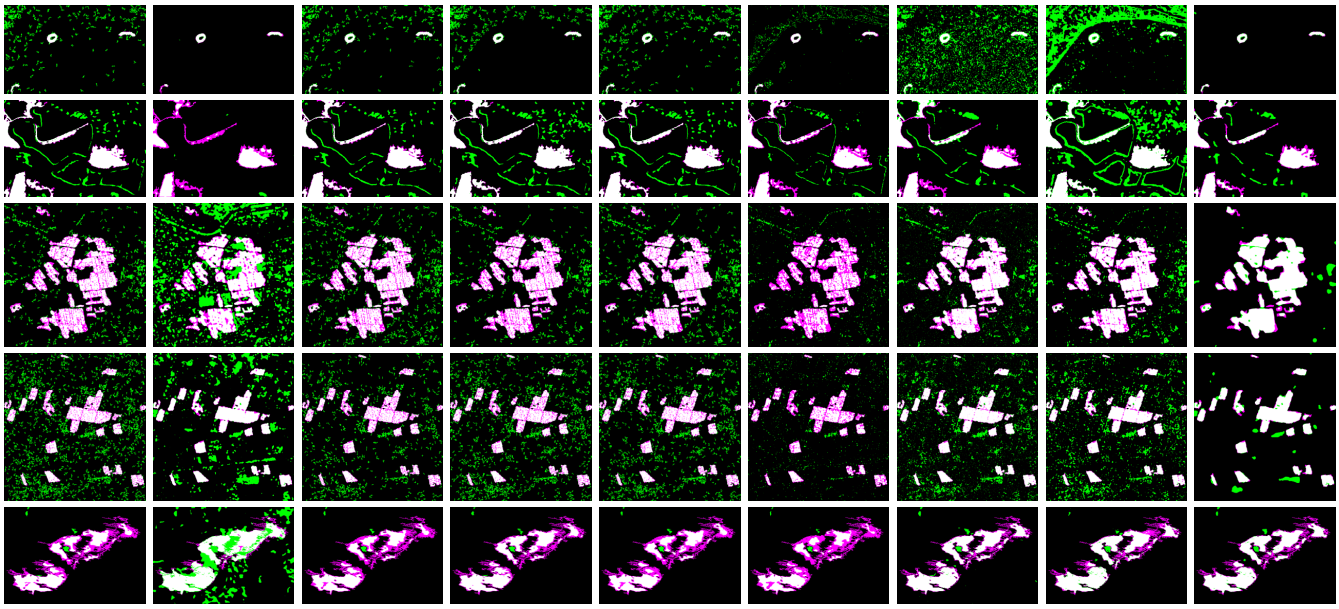


Fig. 11. Visual results obtained by different change detection methods: (a) DDNet; (b) INLPG; (c) LANTNet; (d) CAMixer; (e) WBANet; (f) USSD; (g) M2HG; (h) SWSN; and (i) GRSG-DAF.

TABLE III

QUANTITATIVE EVALUATION OF DIFFERENT CHANGE DETECTION METHODS ON DATASET #2. BEST RESULTS ARE BOLDED, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Method	FP	FN	OE	PCC(%)	KC(%)	Time(s)
DDNet [11]	6,045	1,408	7,453	93.22	73.24	400.49
INLPG [57]	**244**	6,692	6,936	93.69	64.58	19.24
LANTNet [25]	5,130	1,932	7,062	93.58	73.63	433.74
CAMixer [45]	7,757	853	8,610	92.17	70.80	67.74
WBANet [46]	6,309	1,410	7,719	92.98	72.48	53.44
USSD [63]	3,133	4,620	7,753	92.95	66.70	**7.61**
M2HG [28]	3,800	3,880	7,680	93.02	68.43	680.78
SWSN [49]	17,370	**260**	17,630	83.97	52.55	75.21
GRSG-DAF	2,470	3,136	**5,606**	**94.90**	**76.54**	18.98

TABLE IV

QUANTITATIVE EVALUATION OF DIFFERENT CHANGE DETECTION METHODS ON DATASET #3. BEST RESULTS ARE BOLDED, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Method	FP	FN	OE	PCC(%)	KC(%)	Time(s)
DDNet [11]	14,906	17,591	32,497	89.43	65.34	796.29
INLPG [57]	52,964	13,512	66,476	78.38	44.47	125.67
LANTNet [25]	19,838	16,484	36,322	88.19	62.74	982.91
CAMixer [45]	11,996	17,987	29,983	90.25	67.07	154.16
WBANet [46]	15,677	15,798	31,475	89.76	66.99	277.75
USSD [63]	6,044	27,748	33,792	89.01	58.82	**18.79**
M2HG [28]	13,520	11,254	24,774	91.94	74.41	2396.09
SWSN [49]	12,277	11,912	24,189	92.13	74.71	726.19
GRSG-DAF	**5,150**	**6,843**	**11,993**	**96.10**	**87.29**	110.90

to the small number of actually changed pixels, their KC values remain at a relatively low level. Both INLPG and the proposed GRSG-DAF perform excellently in terms of KC values. However, INLPG produces some FNs in the detailed regions of certain changed areas and has a longer runtime than GRSG-DAF. In contrast, the proposed GRSG-DAF achieves the highest PCC and KC values as well as the shortest runtime, demonstrating its superiority.

2) Results on Dataset #2: The changes in Dataset #2 mainly occur in water regions, where the presence of noise results in some FPs, posing a challenge for change detection. The second row of Fig. 11 and Table III displays the performance evaluation results of various change detection methods applied to Dataset #2. It is noticeable that INLPG and USSD tend to produce more FNs, while the detection errors of other methods are mainly attributed to an increase in FPs. Additionally, USSD achieves the shortest runtime with its block-processing strategy, but its accuracy is lower than that of most learning-based methods. Excluding the proposed GRSG-DAF, the top four methods, which are all learning-based, include DDNet, LANTNet, CAMixer, and WBANet, each achieving a KC value above 70%. However, these learning-based methods require significantly longer runtimes due to pixel-based processing. In comparison, the proposed GRSG-DAF reaches the highest KC value of 76.54%, reflecting its strong capability in extracting change features even under noisy conditions.

3) Results on Dataset #3: In the third row of Fig. 11 and Table IV, the visual representations and quantitative evaluations for different methods on Dataset #3 are presented. Dataset #3 mainly highlights changes in earthquake-affected regions, where the concentrated changed areas tend to result in a certain amount of FNs. Meanwhile, due to the complex noise in the unchanged areas, some FPs are also introduced, posing challenges for change detection. It can be observed that several methods, including M2HG and SWSN, provide moderate accuracy but at higher computational costs, while INLPG and USSD fall short in balancing efficiency and performance. In contrast, the proposed GRSG-DAF achieves

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TABLE V
QUANTITATIVE EVALUATION OF DIFFERENT CHANGE DETECTION METHODS ON DATASET #4. BEST RESULTS ARE BOLDED, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Method	FP	FN	OE	PCC(%)	KC(%)	Time(s)
DDNet [11]	43,815	8,495	52,310	86.69	46.47	660.28
INLPG [57]	32,741	**3,788**	36,529	90.70	60.55	203.28
LANTNet [25]	23,330	9,846	33,176	91.56	58.61	1017.18
CAMixer [45]	35,809	9,179	44,988	88.55	50.33	156.38
WBANet [46]	32,121	9,226	41,347	89.48	52.87	355.23
USSD [63]	7,997	14,290	22,287	94.33	65.31	**30.29**
M2HG [28]	32,124	3,927	36,051	90.83	60.81	3524.52
SWSN [49]	31,630	4,270	35,900	90.86	60.69	1290.91
GRSG-DAF	**7,167**	3,984	**11,151**	**97.16**	**84.49**	140.89

TABLE VI
QUANTITATIVE EVALUATION OF DIFFERENT CHANGE DETECTION METHODS ON DATASET #5. BEST RESULTS ARE BOLDED, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Method	FP	FN	OE	PCC(%)	KC(%)	Time(s)
DDNet [11]	506	11,150	11,656	91.24	62.39	275.22
INLPG [57]	14,897	**2,763**	17,660	86.73	61.54	27.13
LANTNet [25]	430	11,671	12,101	90.90	60.46	506.99
CAMixer [45]	495	8,968	9,463	92.89	70.67	30.27
WBANet [46]	418	11,832	12,250	90.79	59.82	36.85
USSD [63]	**350**	13,897	14,247	89.29	51.03	10.63
M2HG [28]	903	6,610	7,513	94.35	78.05	719.55
SWSN [49]	966	6,721	7,687	94.22	77.52	138.48
GRSG-DAF	935	6,354	**7,289**	**94.52**	**78.82**	**7.68**

the best performance across all five accuracy metrics while maintaining competitive runtime.

4) Results on Dataset #4: The fourth row of Fig. 11 and Table V shows the visual and quantitative evaluation results of different methods applied to Dataset #4, respectively. Similar to Dataset #3, Dataset #4 exhibits changes in earthquake-affected regions. Due to the complexity of the background, various methods produce FPs similar to those observed in Dataset #3. These FPs are mainly caused by noise in unchanged areas. Notably, learning-based methods, including DDNet, LANTNet, CAMixer, WBANet, and SWSN, tend to have more FNs and longer runtimes. This trend reflects their limitations in capturing global neighborhood features. Although USSD achieves the shortest runtime, its bilateral filtering approach tends to suppress change details, leading to the highest FNs and relatively lower overall accuracy. Conversely, the proposed GRSG-DAF performs better than other methods in terms of FP, OE, PCC, and KC, validating its superiority both visually and quantitatively.

5) Results on Dataset #5: The changes in Dataset #5 are also caused by earthquake activity. The fifth row of Fig. 11 and Table VI presents the evaluation results of various change detection methods applied to Dataset #5. Although the dataset is relatively small, the short time interval (only one month) between the bitemporal SAR images makes some changed areas difficult to detect. From the visual results of each method, it is noticeable that FNs have a greater impact on the detection results compared to the previous four datasets. Moreover, INLPG exhibits a large number of FPs, 14 897,

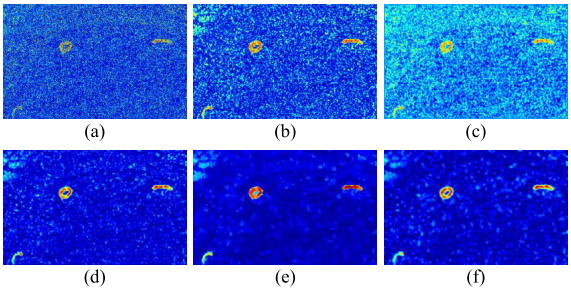


Fig. 12. DIs of different methods applied to Dataset #1: (a) LR; (b) MR; (c) NR; (d) SLRDI; (e) MSRDI; and (f) GRSG-DAF.

resulting in lower PCC values. M2HG performs well with a KC value of 78.05%, but its pixelwise graph construction results in the longest runtime. Even though the proposed GRSG-DAF is not the best in terms of FP and FN, it still achieves the highest PCC and KC while maintaining the shortest runtime, demonstrating its strong ability to balance the detection of changed and unchanged areas while ensuring computational efficiency.

V. ABLATION STUDIES AND DISCUSSION

In this section, the performance of the proposed method will be evaluated from the following perspectives: 1) analysis of different DI generation methods; 2) analysis of different postprocessing methods; 3) ablation of DAGF; 4) effect of the number of edges in the constructed ego graph; 5) effect of the number of superpixels; and 6) analysis of FP and FN metrics.

A. Analysis of Different DI Generation Methods

As mentioned earlier, the proposed GRSG-DAF method follows the change detection process of generating a DI and then performing segmentation. An effective DI should clearly distinguish between changed and unchanged areas. This section provides an analysis of the DIs through qualitative and quantitative comparisons. To demonstrate the superiority of the proposed GRSG-DAF, five state-of-the-art DI generation methods (LR [16], MR [17], NR [18], SLRDI [34], and MSRDI [34]) were selected for comparison.

The DIs obtained from different methods across five datasets are shown in Figs. 12–16. To display the pixel-difference information clearly, all DIs are presented in the form of heatmaps. By analyzing the results of different methods, we observe that LR, MR, and NR produce numerous irregular spots in the DIs across all five datasets. This issue arises from their insufficient suppression of noise under varying conditions. The DIs generated by SLRDI show relatively smooth unchanged areas in smaller datasets (Datasets #1, #2, and #5), whereas the DIs in larger datasets (Datasets #3 and #4) still exhibit numerous speckles and do not prominently highlight the changed areas. Although MSRDI demonstrates good noise resistance, it falls short of capturing boundary structures. In contrast, the proposed GRSG-DAF method effectively suppresses noise and preserves clear boundaries, which is achieved by utilizing global information propagation and DAGF strategies.

The receiver operating characteristic (ROC) and precision–recall (PR) curves of different DI generation

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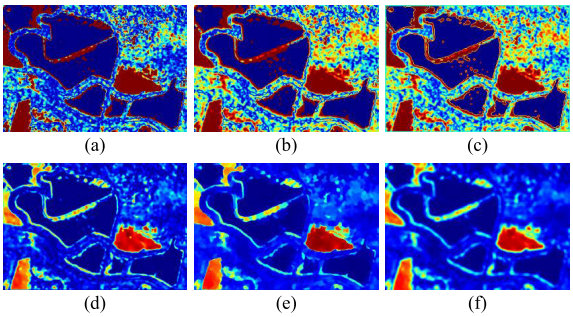


Fig. 13. DIs of different methods applied to Dataset #2: (a) LR; (b) MR; (c) NR; (d) SLRDI; (e) MSRDI; and (f) GRSG-DAF.

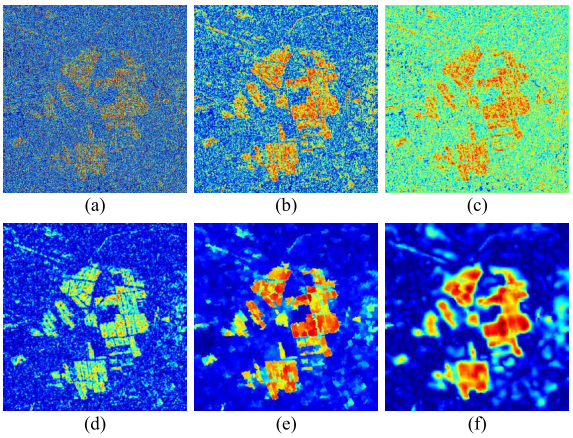


Fig. 14. DIs of different methods applied to Dataset #3: (a) LR; (b) MR; (c) NR; (d) SLRDI; (e) MSRDI; and (f) GRSG-DAF.

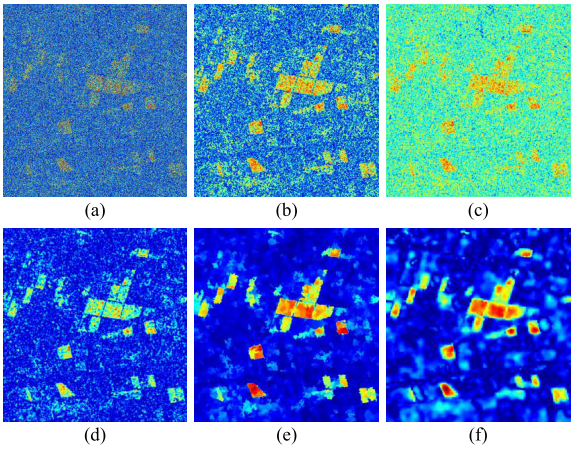


Fig. 15. DIs of different methods applied to Dataset #4: (a) LR; (b) MR; (c) NR; (d) SLRDI; (e) MSRDI; and (f) GRSG-DAF.

methods are plotted in Fig. 17. According to the previous analysis, the proposed GRSG-DAF can more effectively distinguish between changed and unchanged areas. Its ROC curves on Datasets #1, #3, #4, and #5 are closer to the top-left corner compared to other methods. Additionally, its AP curves are the closest to the top-right corner across all five datasets, highlighting its effectiveness. The corresponding AUC and AP values of these curves are listed in Table VII. The proposed GRSG-DAF achieves the highest AUC and AP values on Datasets #1, #3, #4, and #5. For Dataset

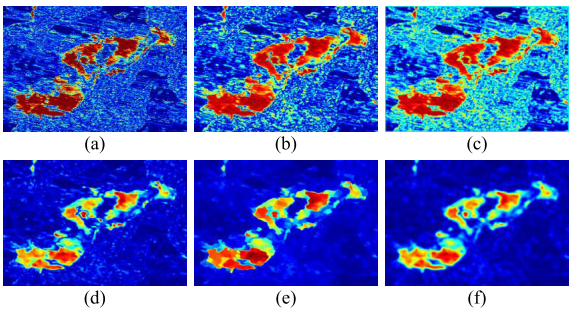


Fig. 16. DIs of different methods applied to Dataset #5: (a) LR; (b) MR; (c) NR; (d) SLRDI; (e) MSRDI; and (f) GRSG-DAF.

#2, the proposed GRSG-DAF attains the highest AP value, while MSRDI achieves the highest AUC value. Nevertheless, MSRDI performs worse than GRSG-DAF on other datasets, demonstrating the robustness of the proposed GRSG-DAF in SAR change detection tasks.

B. Analysis of Different Postprocessing Methods

Given the potential variations in accuracy and computational efficiency among postprocessing methods, a systematic comparison was conducted among five approaches: Otsu thresholding [36], FCM [37], FLICM [38], PCAK [39], and CRF [40]. The results presented in Table VIII indicate that all methods achieve high accuracy in terms of PCC and KC values. This consistent performance implies that the DI generated by GRSG-DAF effectively distinguishes between changed and unchanged areas. Regarding computational efficiency, FLICM exhibits the longest runtime while Otsu achieves the shortest. However, Otsu's accuracy is consistently outperformed by CRF across multiple datasets. Considering the balance between accuracy and computational efficiency, the CRF approach is selected as the postprocessing method used in this study.

C. Ablation of DAGF

To evaluate the contribution of the designed DAGF in SAR change detection tasks, comparative experiments were conducted on five datasets. The results, as shown in Fig. 18, demonstrate the necessity of incorporating DAGF in improving detection accuracy. Specifically, by leveraging gradient information from the original pixel-level image, DAGF enhances detailed retention and ensures more precise change detection. The removal of DAGF results in lower KC values, highlighting its significance in distinguishing subtle changes in SAR images. In summary, the five datasets, which vary in size, texture, and noise levels, provide a rigorous test environment for DAGF, confirming its critical role in enhancing SAR change detection performance.

D. Effect of the Number of Edges in the Constructed Ego Graph

The constructed ego graph for image reconstruction depends on a specific number of edges. Adjusting this number can directly affect the performance of the change detection

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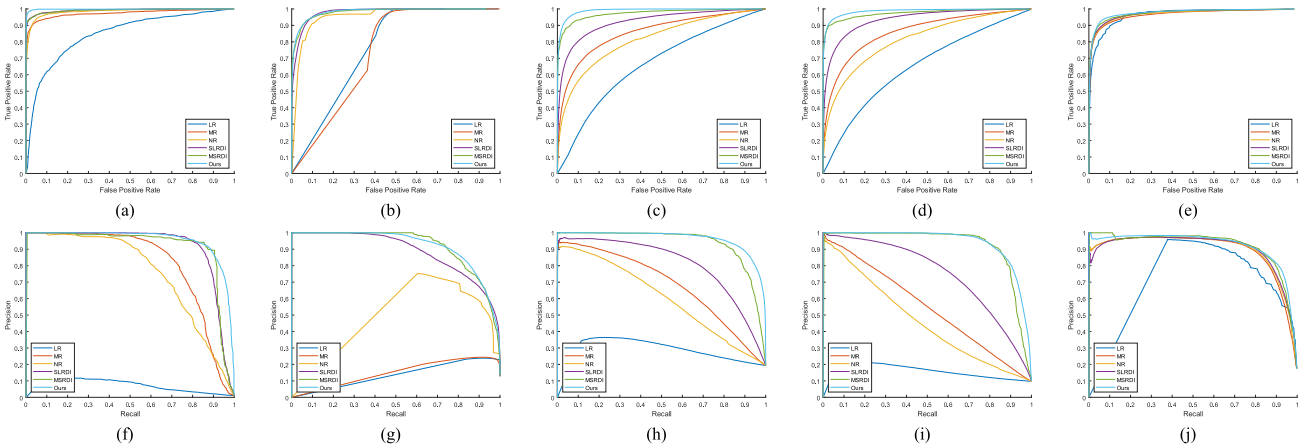


Fig. 17. ROC and PR curves of DIs obtained by various methods. (a)–(e) ROC curves. (f)–(j) PR curves. From left to right, the results correspond to Datasets #1–#5, respectively.

TABLE VII

COMPARISON OF AUC AND AP VALUES OBTAINED BY DIFFERENT DI GENERATION METHODS ON FIVE DATASETS. BEST RESULTS ARE BOLDED, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Dataset	AUC						AP					
	LR	MR	NR	SLRDI	MSRDI	GRSG-DAF	LR	MR	NR	SLRDI	MSRDI	GRSG-DAF
#1	0.8221	0.9670	0.9833	0.9922	0.9910	**0.9982**	0.0702	0.8135	0.7581	0.9144	0.9114	**0.9540**
#2	0.7524	0.7357	0.6542	0.7143	**0.9194**	0.9159	0.1353	0.1480	0.4759	0.8868	0.9185	**0.9198**
#3	0.6511	0.8546	0.8144	0.9226	0.9725	**0.9750**	0.2813	0.6833	0.6040	0.8091	0.9367	**0.9640**
#4	0.6450	0.8577	0.8148	0.9369	0.9735	**0.9759**	0.1578	0.5485	0.4618	0.7468	0.9202	**0.9395**
#5	0.8082	0.9179	0.9320	0.9551	0.9609	**0.9637**	0.6848	0.8927	0.8980	0.9009	0.9189	**0.9228**

TABLE VIII

COMPARISON OF PCC, KC, AND RUNTIME VALUES OBTAINED BY DIFFERENT POSTPROCESSING METHODS ON FIVE DATASETS. BEST RESULTS ARE BOLDED, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Dataset	PCC(%)					KC(%)					Time(s)				
	Otsu	FCM	FLICM	PCAK	CRF	Otsu	FCM	FLICM	PCAK	CRF	Otsu	FCM	FLICM	PCAK	CRF
#1	**99.74**	99.35	99.70	99.71	99.73	**88.08**	76.06	86.07	87.05	87.41	**0.10**	0.74	4.95	0.64	0.23
#2	94.27	94.28	**95.18**	95.11	94.90	74.22	74.19	77.36	**77.58**	76.54	**0.02**	0.12	2.46	0.46	0.08
#3	96.02	96.07	96.06	95.99	**96.10**	86.96	87.09	87.03	86.86	**87.29**	**0.04**	0.30	6.23	1.25	0.18
#4	97.17	97.04	**97.36**	97.19	97.16	84.40	83.89	**84.99**	84.48	84.49	**0.02**	0.58	26.25	1.60	0.21
#5	94.49	94.39	94.37	94.43	**94.52**	78.77	78.56	78.16	78.60	**78.82**	**0.02**	0.17	2.01	0.55	0.11

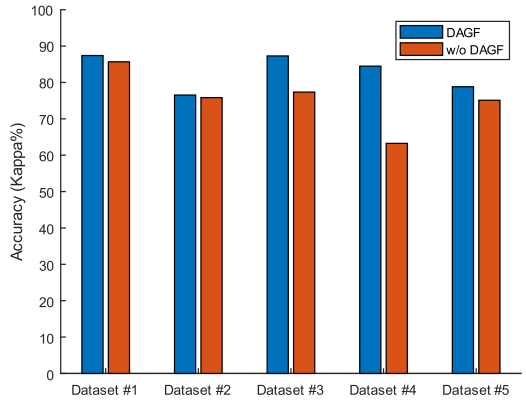


Fig. 18. Comparison of KCs with and without DAGFs across five datasets.

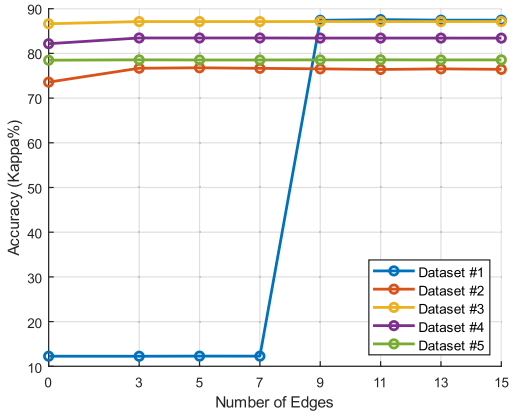


Fig. 19. Comparisons of KCs with different numbers of edges.

process. By setting the number of superpixels to 3000 and varying the number of edges from 0 to 15 (with 0 indicating that no image reconstruction strategy is applied), we evaluate the impact of edge count on KC values, as shown in Fig. 19. For Datasets #2–#5, KC values are lower without image reconstruction. After applying image reconstruction, KC values show no significant fluctuation after the improvement. This is because the aggregated neighborhood information remains largely consistent, regardless of the edge count. For Dataset

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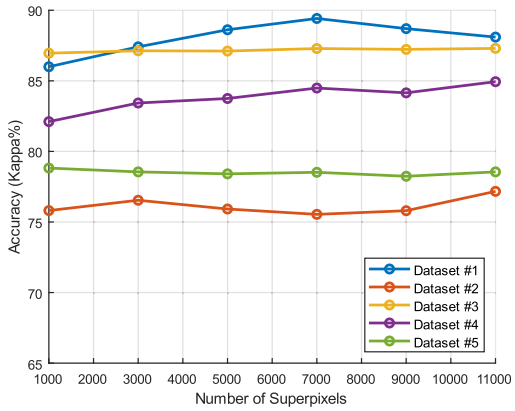


Fig. 20. Comparisons of KCs with different numbers of superpixels.

#1, a significant imbalance exists between the changed and unchanged areas. When the number of edges is fewer than 9, insufficient neighborhood information results in low KC values. Meanwhile, as the number of edges exceeds 9, detection accuracy for changed areas improves significantly, leading to an increase in KC values. Based on these findings, setting the edge count to 9 is recommended for experiments. This experiment also validates the effectiveness of the proposed image reconstruction strategy.

E. Effect of the Number of Superpixels

The number of superpixels determines the granularity of information division in each SAR image, thereby influencing the final change detection results. On the one side, insufficient superpixels may cause change-prone pixels to be included, leading to the loss of critical change information during image reconstruction. On the other side, excessive superpixels may fail to capture the overall change within the same semantic region and impose an increased computational burden. As analyzed above, with the edge count fixed at 9 during reconstruction, the number of superpixels is varied from 1000 to 11 000 to evaluate the performance of the proposed GRSG-DAF method. Fig. 20 shows the relationship between the number of superpixels and KC values across the five datasets. Due to the varying sizes of the datasets, the trends of the curves differ. Considering both accuracy and computational load, the number of superpixels for Datasets #1, #2, and #5 can be kept under 3000, while for Datasets #3 and #4, it can be kept under 7000.

F. Analysis of FP and FN Metrics

A detailed examination of FP and FN values is essential to understanding how well the model distinguishes between changed and unchanged areas. In this analysis, we focus on graph construction parameters since adjusting the number of edges and superpixels helps reveal how structural information propagation affects misclassification. Figs. 21 and 22, respectively, show the relationship between the number of edges and FP/FN, as well as the relationship between the number of superpixels and FP/FN. Fig. 21 demonstrates

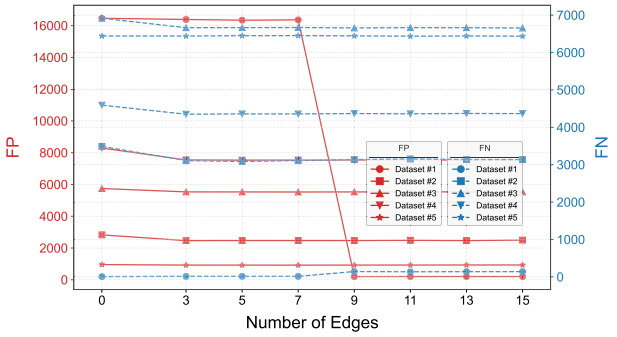


Fig. 21. Relationship between edge count and FP/FN.

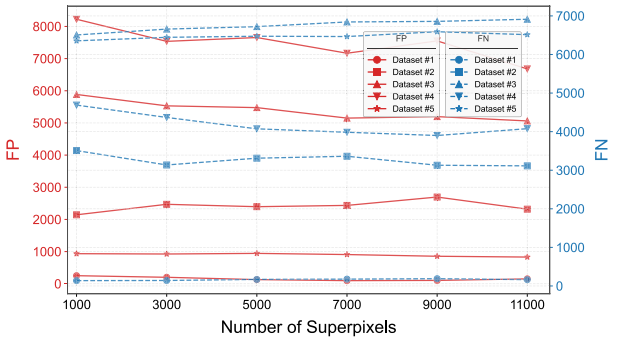


Fig. 22. Relationship between superpixel count and FP/FN.

that a reduced edge count typically tends to increase FP due to insufficient contextual integration, while an increased edge count suppresses FP at the expense of raising FN by over-smoothing subtle changes. Similarly, Fig. 22 illustrates that employing a larger number of superpixels improves local detailed preservation (thereby reducing FP), while an excessive superpixel count disrupts coherent change regions, leading to an increase in FN.

VI. Conclusion

This article focuses on graph theory-based change detection in SAR images. We begin by reviewing previous approaches to change detection in SAR images based on structured graphs. These methods often suffer from missing contextual and spatial details, as well as high computational complexity, which negatively affects their performance. To address these challenges, we propose the GRSG-DAF method, which integrates difference information more effectively. Specifically, we first employ a difference-aware strategy to segment the image into superpixels and construct a structured graph. This graph captures the relationships and similarities among superpixel vertices, helping to preserve the structural information of the image while reducing computational complexity. Additionally, we propose an image reconstruction process based on a feature propagation mechanism within the structured graph, allowing for more effective integration of global features. Finally, we design a DAGF process, which combines information at both the pixel and superpixel levels to improve the accuracy of change detection. The experimental results on five SAR change

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detection datasets indicate the excellence of our method compared to other related methods.

While GRSG-DAF demonstrates strong performance across various datasets, its performance can be influenced by hyperparameter settings in scenarios with extremely imbalanced class distributions. Proper parameter selection can improve detection performance, whereas inappropriate settings may make it difficult for the model to effectively distinguish between changed and unchanged areas. Although GRSG-DAF optimizes overall detection performance, there is still room for improvement in detecting changed and unchanged areas individually. Future research could explore more robust feature-learning strategies to reduce dependence on hyperparameters and improve the model's adaptability and stability across different scenarios.

APPENDIX
DERIVATION OF GUIDED FILTERING COEFFICIENTS

In this appendix, we derive the parameters $a_{\mathbf{x}}$ and $b_{\mathbf{x}}$ in the DAGF using image $\boldsymbol{X}$ as an example.

Assume that the filtered image is a linear transformation of the guide image $\boldsymbol{X}$ within a certain window

$$f_X(\mathbf{x}) = a_{\mathbf{x}}\boldsymbol{X}(x') + b_{\mathbf{x}} \ \forall x' \in \Omega_{\mathbf{x}}. \tag{A.1}$$

Based on the constructed loss function

$$\mathcal{J}(a_{\mathbf{x}}, b_{\mathbf{x}}) = \sum_{x'\in\Omega_{\mathbf{x}}}\left(\left(a_{\mathbf{x}}\boldsymbol{X}(x') + b_{\mathbf{x}} - \boldsymbol{P}_{\boldsymbol{GX}}(x')\right)^2 + \frac{\epsilon}{\Gamma_{\mathrm{IDI}}(x')}a_{\mathbf{x}}^2\right) \tag{A.2}$$

we take the derivative of both sides of (A.2) with respect to $a_{\mathbf{x}}$ and $b_{\mathbf{x}}$

$$\frac{\partial}{\partial a_{\mathbf{x}}}\mathcal{J}(a_{\mathbf{x}}, b_{\mathbf{x}}) = \sum_{x'\in\Omega_{\mathbf{x}}}\left(\left(a_{\mathbf{x}}\boldsymbol{X}(x') + b_{\mathbf{x}} - \boldsymbol{P}_{\boldsymbol{GX}}(x')\right)\boldsymbol{X}(x') + \frac{\epsilon}{\Gamma_{\mathrm{IDI}}(x')}a_{\mathbf{x}}\right) \tag{A.3}$$

$$\frac{\partial}{\partial b_{\mathbf{x}}}\mathcal{J}(a_{\mathbf{x}}, b_{\mathbf{x}}) = \sum_{x'\in\Omega_{\mathbf{x}}}\left(\left(a_{\mathbf{x}}\boldsymbol{X}(x') + b_{\mathbf{x}} - \boldsymbol{P}_{\boldsymbol{GX}}(x')\right)\right) = a_{\mathbf{x}}\sum_{x'\in\Omega_{\mathbf{x}}}\boldsymbol{X}(x') + b_{\mathbf{x}}\sum_{x'\in\Omega_{\mathbf{x}}}1 - \sum_{x'\in\Omega_{\mathbf{x}}}\boldsymbol{P}_{\boldsymbol{GX}}(x'). \tag{A.4}$$

By setting (A.4) to zero, we can solve for $b_{\mathbf{x}}$

$$b_{\mathbf{x}} = \frac{1}{|\Omega_{\mathbf{x}}|}\sum_{x'\in\Omega_{\mathbf{x}}}\boldsymbol{P}_{\boldsymbol{GX}}(x') - a_{\mathbf{x}}\left(\frac{1}{|\Omega_{\mathbf{x}}|}\sum_{x'\in\Omega_{\mathbf{x}}}\boldsymbol{X}(x')\right) = E_{\Omega_{\mathbf{x}}}[\boldsymbol{P}_{\boldsymbol{GX}}(\mathbf{x})] - a_{\mathbf{x}}E_{\Omega_{\mathbf{x}}}[\boldsymbol{X}(\mathbf{x})]. \tag{A.5}$$

Substituting the obtained $b_{\mathbf{x}}$ into (A.3) yields

$$\frac{\partial}{\partial a_{\mathbf{x}}}\mathcal{J}(a_{\mathbf{x}}, b_{\mathbf{x}}) = \sum_{x'\in\Omega_{\mathbf{x}}}\Bigg(\left(a_{\mathbf{x}}\boldsymbol{X}(x') + E_{\Omega_{\mathbf{x}}}[\boldsymbol{P}_{GX}(\mathbf{x})] - a_{\mathbf{x}}E_{\Omega_{\mathbf{x}}}[\boldsymbol{X}(\mathbf{x})]\right.$$
$$\left.-\boldsymbol{P}_{GX}(x')\right)\boldsymbol{X}(x') + \frac{\epsilon}{\Gamma_{\mathrm{IDI}}(x')}a_{\mathbf{x}}\Bigg)$$
$$= \sum_{x'\in\Omega_{\mathbf{x}}}\Bigg(a_{\mathbf{x}}\left(\boldsymbol{X}(x')\boldsymbol{X}(x') - \boldsymbol{X}(x')E_{\Omega_{\mathbf{x}}}[\boldsymbol{X}(\mathbf{x})] + \frac{\epsilon}{\Gamma_{\mathrm{IDI}}(x')}\right) + \boldsymbol{X}(x')E_{\Omega_{\mathbf{x}}}[\boldsymbol{P}_{\boldsymbol{GX}}(\mathbf{x})] - \boldsymbol{P}_{\boldsymbol{GX}}(x')\boldsymbol{X}(x')\Bigg). \tag{A.6}$$

Equation (A.6) is set to zero, and it follows that:

$$a_{\mathbf{x}}\Bigg(\sum_{x'\in\Omega_{\mathbf{x}}}\left(\boldsymbol{X}(x')\boldsymbol{X}(x')\right) - E_{\Omega_{\mathbf{x}}}[\boldsymbol{X}(\mathbf{x})]\sum_{x'\in\Omega_{\mathbf{x}}}\left(\boldsymbol{X}(x')\right) + \epsilon\sum_{x'\in\Omega_{\mathbf{x}}}\left(\frac{1}{\Gamma_{\mathrm{IDI}}(x')}\right)\Bigg) = \sum_{x'\in\Omega_{\mathbf{x}}}\left(\boldsymbol{P}_{\boldsymbol{GX}}(x')\boldsymbol{X}(x')\right) - E_{\Omega_{\mathbf{x}}}[\boldsymbol{P}_{\boldsymbol{GX}}(\mathbf{x})]\sum_{x'\in\Omega_{\mathbf{x}}}\left(\boldsymbol{X}(x')\right). \tag{A.7}$$

Multiply both sides of (A.7) by $(1/|\Omega_{\mathbf{x}}|)$ and simplify to obtain

$$a_{\mathbf{x}}\left(E_{\Omega_{\mathbf{x}}}[\boldsymbol{X}(\mathbf{x})\boldsymbol{X}(\mathbf{x})] - E_{\Omega_{\mathbf{x}}}[\boldsymbol{X}(\mathbf{x})]E_{\Omega_{\mathbf{x}}}[\boldsymbol{X}(\mathbf{x})] + E_{\Omega_{\mathbf{x}}}\left[\epsilon/\Gamma_{\mathrm{IDI}}(\mathbf{x})\right]\right) = E_{\Omega_{\mathbf{x}}}[\boldsymbol{X}(\mathbf{x})\boldsymbol{P}_{\boldsymbol{GX}}(\mathbf{x})] - E_{\Omega_{\mathbf{x}}}[\boldsymbol{X}(\mathbf{x})]E_{\Omega_{\mathbf{x}}}[\boldsymbol{P}_{\boldsymbol{GX}}(\mathbf{x})]. \tag{A.8}$$

According to the formulas for variance and covariance, $\mathrm{Var}[X] = E[X^2] - E[X]^2$, $\mathrm{Cov}[X, Y] = E[XY] - E[X]E[Y]$, we finally have

$$a_{\mathbf{x}} = \frac{\mathrm{Cov}_{\Omega_{\mathbf{x}}}[\boldsymbol{X}(\mathbf{x}), \boldsymbol{P}_{\boldsymbol{GX}}(\mathbf{x})]}{\mathrm{Var}_{\Omega_{\mathbf{x}}}[\boldsymbol{X}(\mathbf{x})] + E_{\Omega_{\mathbf{x}}}\left[\epsilon/\Gamma_{\mathrm{IDI}}(\mathbf{x})\right]}. \tag{A.9}$$

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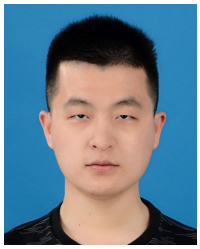
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